Feature

16-Bit MS-DOS & CP/M For Your Osborne One!

The CO-POWER-88 board adds an 8088 processor, the MS-DOS, CP/M-86 and CP/M systems, and a 256K memory disk available under regular CP/M.

Robert J. Thomson

Your eyes do not deceive you. My Osborne 1 has been modified to run 16-bit software on both the MS-DOS™ and CP/M-86™ operating systems, while still retaining the capability to run all of my CP/M® 8-bit software. Not only that! I also have a 252K RAM disk available when running in the 8-bit mode. And all for only $560!!

This magic is performed by the installation of CO-POWER-88 from SWP Microcomputer Products, Inc. of Arlington, Texas. CO-POWER-88 is an add-on board with an 8088 microprocessor chip and some memory chips that provide 256K of RAM and the software to drive it.

I first talked to SWP about a version of CO-POWER-88 for the Osborne 1 in mid-December. It was ordered in mid-February. It arrived in mid-April. I seem to have a mid-month fetish. No matter. It is now installed, and has been running for a month, and works fine!

Like so many companies in the microcomputer industry, Software Publishers (SWP) seem to be having ordering and delivery problems. When you call to enquire about your order, they are "waiting for parts" or some such response, and they are expecting to ship "real soon." There is a hint that they have been swamped with orders and are trying, valiantly, to catch up. Anyway, if you expect about a two-month wait for your order to arrive you probably wont be disappointed. If it shows up earlier than expected, you will be pleased; if not, it's what you expected.

CO-POWER-88 has been available for the Kaypro II computer for some time but only recently has it become available in an Osborne 1 configuration. The principal differences between the Osborne and Kaypro configurations are in the hardware pieces which adapt the board to the computer chassis, and the power adapter cable. The 8088 board itself is the Kaypro board — it says so right on it!

The notes provided by SWP state that this installation describes the interface with Osborne 1s which have undergone the 80 column (SCREENPAC) upgrade. All instructions refer to how it interfaces with the Osborne's 80 column board. Osborne 1s that have this upgrade are, of course, double-density machines.

The installation is straightforward, requiring no soldering. However, since the board is installed inside the case, you must disassemble your Osborne. If you have the blue case Osborne 1, this is easy to do. But I have one of the old tan case machines.

What You Get

The number of separate parts provided is small. The kit, shown in figure 1, contains the following hardware:

- Main 8088 board, which is approximately $5^3/_4$" by 7", contains the 8088 microprocessor, 256K of memory and several other support chips.

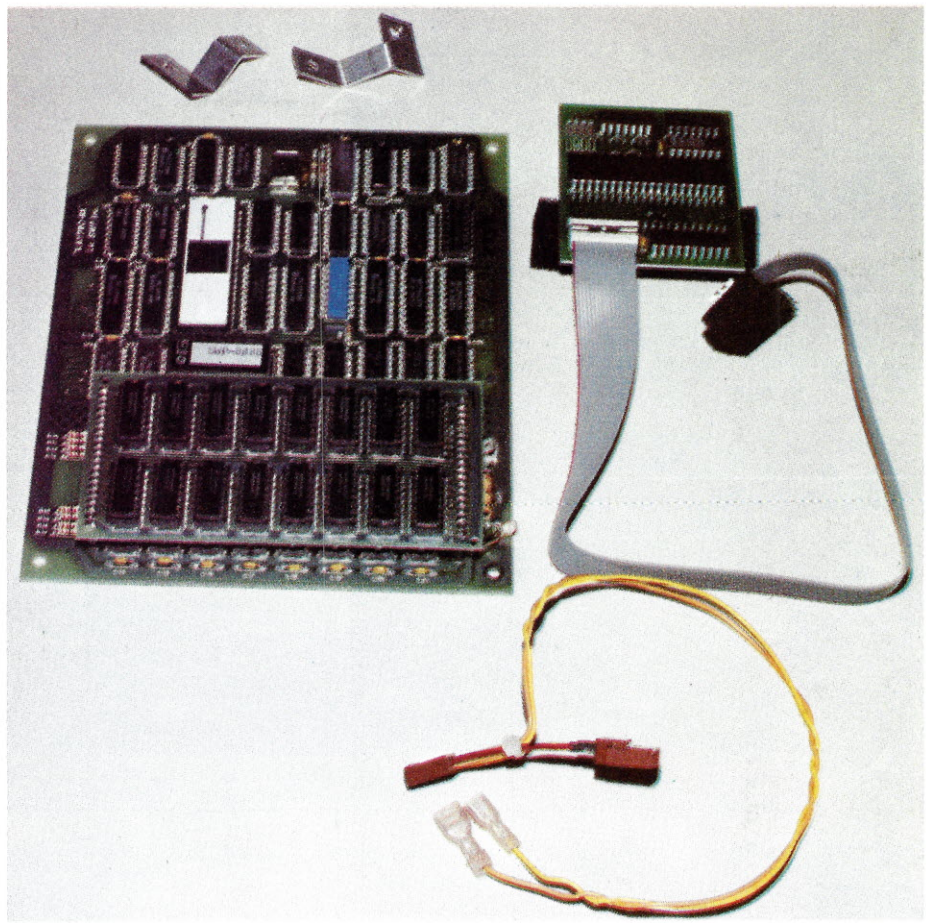
- Adapter board, about 3" by $2^1/_2$" containing a Z80A processor, two other support chips and 13 inches of flat 16 conductor cable. The connector at the end of this cable fits into a receptacle on the 8088 board during installation.

- Power cable assembly.

Plus you get a package of installation hardware including:

- Two aluminum "Z"-shaped mounting brackets with 4-40 nuts pressed in place.
- Two 4-40 pan head screws.
- A plastic PC board standoff and a metal PC board standoff which screws into the plastic one.

The software provided with the system includes a CP/M-80 disk with programs to load MS-DOS and to configure the RAM Disk, and the MS-DOS version 2.11 operating system. The MS-DOS 2.0 manual is provided along with about ten pages of notes from SWP on how to run their system. CP/M-86 is available as an option. Disks containing the CP/M-86 system and RAM Disk software are provided. The CP/M-86 manual provided is the IBM PC version.



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Figure 1. *The CO-POWER-88 kit from SWP Microcomputer Products adds an 8088 processor to the Osborne 1 or Kaypro II computer.*

Differences: Tan vs. Blue Case

Some of the tan case Osborne 1 components are shaped differently than their counterparts in the blue case, and the Osborne main board mounting arrangement is also different. In both cases, it is necessary to disconnect the keyboard cable, undo and pull off the brightness and contrast control knobs and remove a half dozen or so screws to remove the bezel from the front of the machine.

One major difference between the two Osborne case configurations is the way you gain access to the Osborne 1 main board. The blue case is designed as a two piece unit with the split along the horizontal center line. By turning the machine upside down and unfastening a few screws, the bottom of the case can be removed to reveal the main board. The other half of the case remains in place.

To disassemble the tan case, it is necessary to remove all of the case mounting screws including those holding the carrying handle to the case, and slide the computer forward and out of the case. The tan case remains in one piece. Although the cable assembly of the rear-mounted power switch panel is long enough to allow the machine to slide clear of the case, removal of the few screws holding it in place will allow the computer to be completely separated from the case, greatly simplifying handling of the unit.

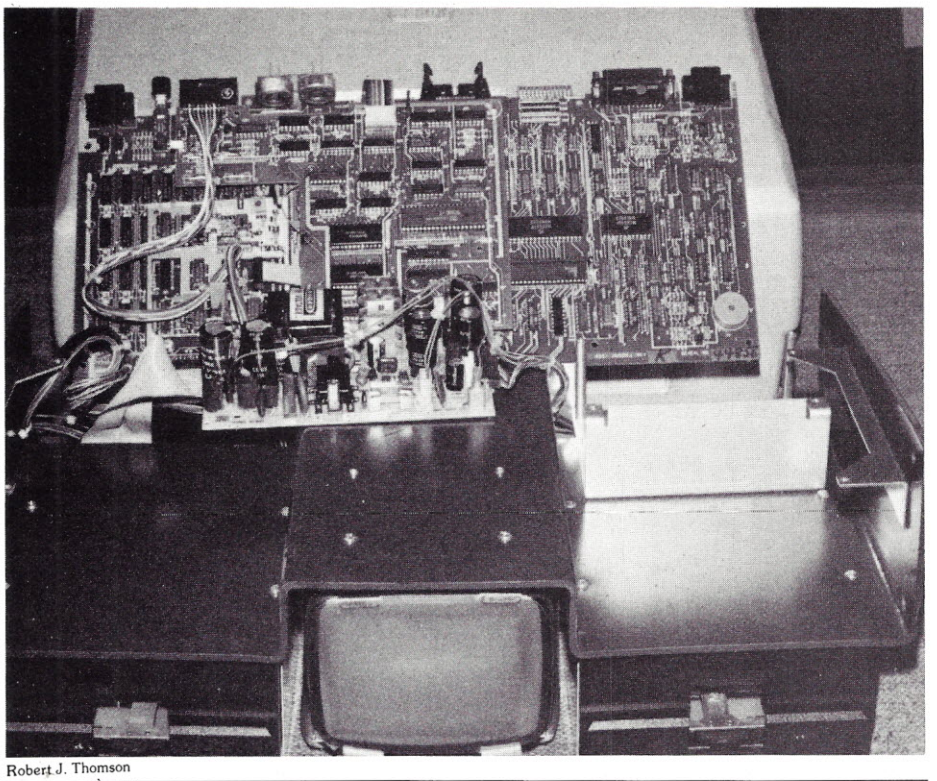
In the blue case, the Osborne main board is mounted to "I"-shaped metal strips which in turn mount on four posts molded into the case. These posts accept the screws which hold the two halves of the case together. The "Z" brackets provided to mount the 8088 board are designed to mount on the two posts on the A drive side. When mounted this way, the 8088 board will not interfere with the Z80 adapter board which replaces the Osborne Z80.

In the tan case Osborne, a metal angle bracket is pop-riveted to the plastic chassis providing mounting support for the Osborne main board (see figure 2). Consequently, while the aluminum "Z" brackets provided by SWP to mount the 8088 board in the Osborne may work fine with the blue case, they need modification and different placement for use with the tan case.

This is not a significant problem, but it does require modification of the "Z" brackets. Two holes must be drilled in the existing metal angle bracket to accept the "Z" brackets provided to mount the 8088 board, as shown in figure 3.

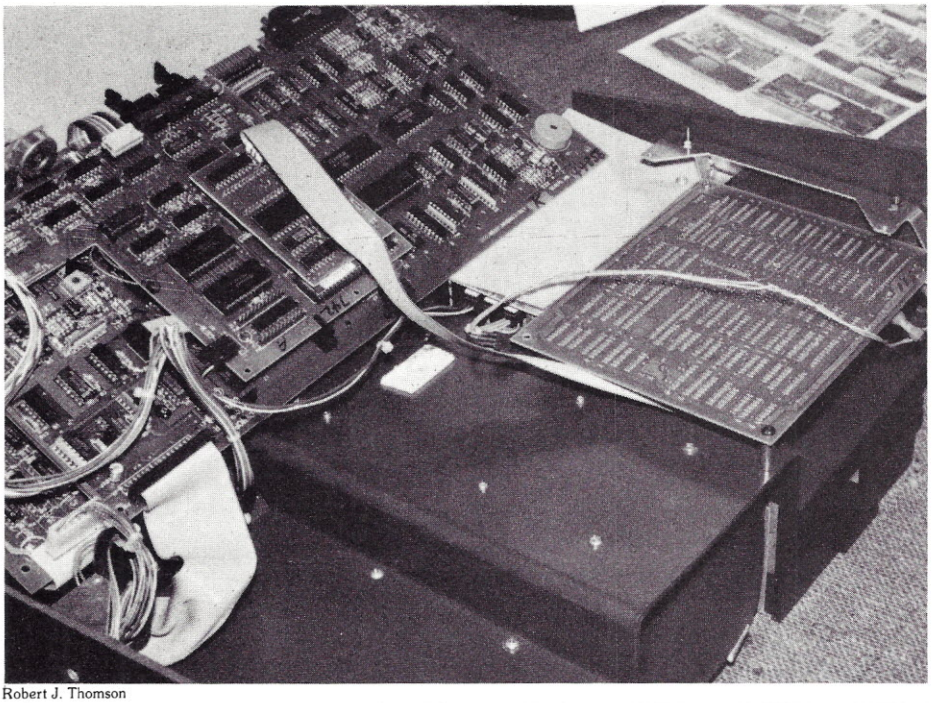
Relocating the 8088 Board

Facing the Osborne 1 CRT, the 8088 board is located on the left side, just under the A drive and above the main board. Since the bezel contains two "disk pockets" which fit between the disk drives and the main board, the 8088 board has to be installed close to the Osborne main board to avoid interference with the disk pockets in the bezel



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Figure 2. *In the tan case Osborne, a metal angle bracket provides mounting support for the main board.*



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Figure 3. *For tan case Osbornes you must modify the "Z" brackets supplied to support the CO-POWER-88 board. Two holes must be drilled in the existing metal angle bracket shown in figure 2.*

during re-assembly.

In tan case machines, these disk pockets are tapered more than on the blue case models so that the opening is larger at the bezel face. This additional taper requires that the 8088 board be mounted closer to the Osborne main board than proposed by SWP for the blue case machines. Consequently, in the tan case Osborne, instead of the 8088 board being mounted on the "Z" brackets as instructed by SWP, it is necessary to relocate it to the under side of the brackets and to separate it from the brackets with nylon spacers (approximately 1/16th of an inch). The result is that the 8088 board is moved closer to the Osborne main board by about 1/8th of an inch.

Modifying the "Z" Brackets

Modification of the aluminum "Z" brackets required some notching, cropping and re-drilling to fit the 8088 board as close to the outside of the chassis as possible, and to lower it somewhat more than the blue case installation called for. This left sufficient clearance for re-installation of the bezel with its larger disk pockets.

We did the following to modify the "Z" brackets:

1. Interchanged the intended placement of "Z" brackets. That is, the "front" bracket went to the back position and the "back" bracket went to the front position.

2. Notched the rear "Z" bracket to clear the pop rivet which holds the Osborne main board mounting bracket to the plastic chassis frame.

4. Drilled two holes in the Osborne main board mounting bracket to accept the "Z" brackets.

5. Used two 6-32 machine screws, lock washers and nuts to mount the "Z" brackets to the Osborne chassis.

6. Moved the mounting holes for 8088 board on the "Z" brackets outward approximately 1/4" to relieve interference between the inboard edge of 8088 board and the Z80 adapter board, and cropped the bracket to remove the existing mounting hole, so as to assure no interference with the power leads on the 8088 board.

7. Placed nylon washers (spacers) between the board and "Z" brackets and mounted the board on the side of the "Z" brackets closest to

the Osborne board. This required that the front bracket be notched to accept the 8088 board.

8. Installed the nylon and metal PC board standoffs on the front corner of the 8088 board opposite the "Z" brackets. The standoffs provide spacing and support from the Osborne mother board.

The installation after these modifications is shown in figure 4.

Moving the Power Supply Bracket

In my installation, the 8088 board stuck out about 3/32" beyond the front of the plastic chassis to which all Osborne components are mounted, even when the 8088 board was butted against the power supply mounting bracket at the rear of the space. Normally, the bezel mounts flush against this plastic chassis, so unless corrected, the 8088 board would prevent proper flush replacement of the bezel on my tan case machine. The fix for this problem was simple, although before proceeding with it some thought was given to the consequences.

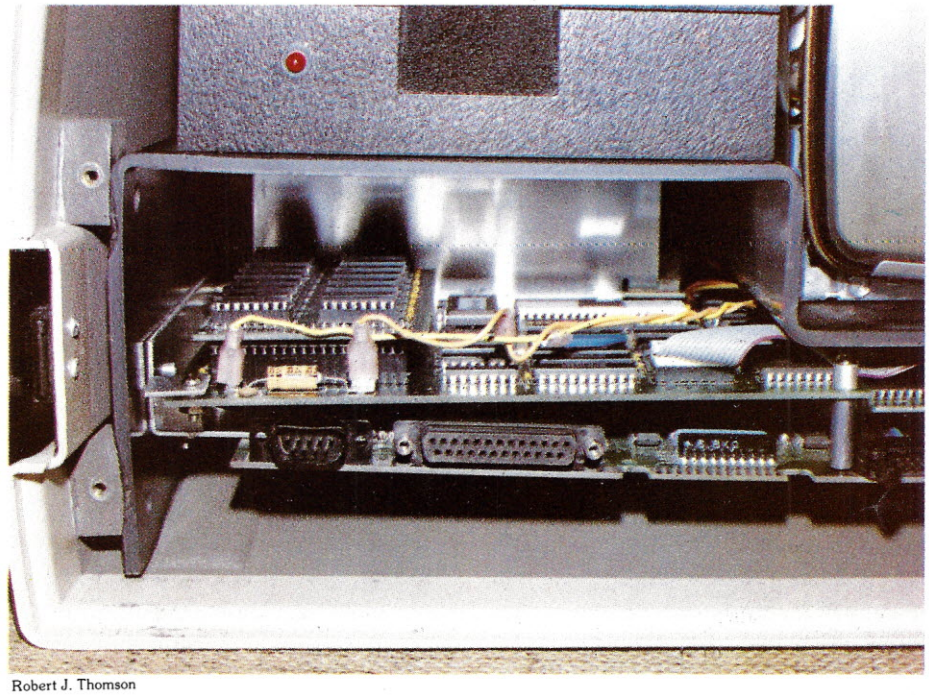
The solution is to remove the Osborne power supply from its mounting bracket (by removing four screws), as shown in figure 2, and to simply bend the mounting bracket and the two mounting posts toward the back about 1/8", sufficient to move the front edge of the 8088 board back until it is flush with the front edge of the tan case Osborne's plastic chassis. After this operation, re-installation of the power supply was achieved with no problems.

Installation of the Z80 Adapter Board

The Z80 adapter board holds the replacement Z80 with the circuitry, interfacing cable and connector to connect it to the 8088 board. The installation of this small board is a simple operation. The most difficult part is removing the existing Z80.

If it is a tight fit, it takes some persistent and careful prying, taking special care not to put stress into the PC board. The adapter board simply plugs into the Z80 receptacle. You must make sure, of course, that the new Z80 is oriented in the same way as the one it is replacing. That is, Pin 1 on the new one is where Pin 1 on the old one was.

There is a 13 inch long ribbon cable to join the adapter board to the 8088 board. A connector on one end plugs into a DIP socket on the Z80 adapter board. The connector on the opposite end plugs into a DIP socket on the 8088 board. Again, its orientation is important, but SWP's instructions are clear and no problems were encountered in this part of the installation. The installed Z80 Adapter Board is shown in figure 3.



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Figure 4. *The resulting installation after modifying the brackets.*

The installation is straightforward, requiring no soldering, but the board must be installed inside the case, and you must disassemble your Osborne.

Power Adapter Cable

My Osborne has the double density and 80 column (SCREENPAC) upgrades which were performed at the Osborne factory. SWP's installation instructions note that they have color coded the power adapter cable to match the (blue case) installation which they used as their model. The color coding matches the upgraded tan case Osborne exactly, so there was no problem with it.

Installation of the power adapter cord involves the simple removal of an existing plug from a receptacle on the power supply, replacing it with the one provided and plugging the original plug into the adapter cable. The other end of the cable has two differently-sized spade (FASTON) connectors which can be attached to the 8088 board only one way. Thus, bringing power to the 8088 is simple and requires no soldering.

If you had an upgrade other than the one Osborne installed, there may be some difference in the power adapter cable arrangement. However, there are only two wires involved. One is 5 volts DC and the other is ground. The 5 volt line goes to the smaller of the two spade (FASTON) connectors. If you are the least bit technically inclined, you should be able to determine which is which.

Putting It Together

After re-assembly of the power supply, the 8088 board was fastened to the modified "Z" brackets as described above, and the PC board standoffs attached to the opposite front corner of the board. With the Z80 Adapter Board installed the Osborne main board was brought into position to be re-mounted on the chassis. The excess length of the ribbon cable, between the Z80 adapter board and the 8088 board, was taken up by folding it over as the boards are brought together during installation. The completed installation is shown in figure 4.

Re-assembly of the computer was uneventful. It is easier to get the Osborne 1 back into the tan case if you turn it upside down before sliding it back into place. The power switch panel is re-installed first. Then, the power cable assembly is carefully fed into the space under B Drive as the computer is slid into the case. The screws holding the case to the handle assembly, the chassis mounting screws and the front bezel screws were replaced and the the machine was plugged in.

The Smoke Test

When the power was turned on the system ran OK, performing all the advertised functions. After running for fifteen or twenty minutes, the Osborne screen went blank and the machine did not respond to commands. Turning the power switch off and on again did not result in the Osborne's familiar flash of disk drive lights and the sign-on "beep." The fuse on the power switch panel was OK. We disassembled the computer to check the fuse on the power supply. It was OK, too. While the machine was disassembled, we plugged it in. Lo! It came alive properly. Everything worked!

Tentative conclusion: The system runs hotter than it did without CO-POWER-88 and activates a thermal protection device in the Osborne 1 (or overheats something until it goes out of tolerance) and shuts the system down. After a few minutes, the system cools enough to begin working again. Cooling the system with a fan yields a reliable working system. A simple $10 desk fan blowing on the front of the machine suffices. However, the addition of a fan to the case would also do the job. The SWP instructions state that a fan *may* be required. In my experience, one *is* required.

How hot is hot? The 8088 chip runs hot to the touch. Surface temperature was measured with a probe and found to be around 50 degrees C (122 degrees F). Apparently this affects the operation of some component to cause a shutdown. However, as noted, you can re-establish reliable operation simply by using a small cooling fan.

Running MS-DOS on the Osborne 1

Loading MS-DOS is simple. SWP provides an MS-DOS "Load" disk which contains CP/M-80 software which instructs CO-POWER-88 to load the MS-DOS system. A screen message tells you to select either Drive A or Drive B as the drive in which MS-DOS will be loaded. The "Load" disk is removed and a disk in MS-DOS format containing at least the MS-DOS file COMMAND.COM is placed in the drive to be selected. Upon keying in the drive selection, MS-DOS loads.

When first loaded, it asks for the date and time. If you enter the date, every file you create in this session will carry that date when placed in the disk directory. Once the system is loaded, you can remove the MS-DOS system disk, and load application programs and data disks from drives A and B.

When exiting from an MS-DOS application program, the system will expect to find the file COMMAND.COM to reactivate MS-DOS. If you copy this file onto the disk in the drive from which the system was booted, you will save a bit of disk swapping.

Two disks are provided by SWP. One is a CP/M-80 disk containing MSDOS.COM, which is the MS-DOS "Load" file, and RAMDISK.COM. The RAMDISK.COM file is used to configure the CO-POWER-88 as a disk emulator for use with CP/M-80 programs.

The other is an MS-DOS System & Utilities disk containing the MS-DOS command processor file COMMAND.COM and standard MS-DOS utility files. SWP has added a file called Z80.EXE which will exit MS-DOS and return to CP/M-80.

Before proceeding to run an MS-DOS program, I formatted some blank disks in MS-DOS format running the MS-DOS utility program FORMAT.COM. Then by loading an MS-DOS version of a program in Drive A and a blank disk in Drive B, for data files, I was all set to check out operation of the Osborne 1 in the 16-bit world.

CP/M 80 MultiPlan Files Run in MS-DOS

I have a CP/M-80 version of MultiPlan for my Osborne 1. An aggravation with the Osborne version is that it is set up to display only 52 columns on the screen, even when you have the 80 column capability.

I installed the MS-DOS version of MultiPlan on the Osborne with CO-POWER-88. It runs fine with a full 80 column display. I did not take time to install the underline, familiar to Osborne owners of MultiPlan or SuperCalc, as the cell marking cursor. Instead, the cursor is marked with square brackets. In the MS-DOS version, the cursor marking brackets do not occupy the same position as the end characters in the cell, thus obliterating them as they do in some CP/M 80 versions I have seen.

Why, you may ask, do I want an MS-DOS version of MultiPlan when I already have a CP/M-80 version? The answer is that in MS-DOS I have 256K of memory available to me, while in CP/M-80 I have only 64K. This can make a tremendous difference in operation, since it allows at least four times as much data to be put into the spreadsheet before encountering the dreaded MEMORY FULL message — and I can display 80 columns of the spreadsheet.

I have just had time to run a few simple tests on MS-DOS, using MultiPlan as the example. I have found that the a MultiPlan spreadsheet created on my Osborne in the CP/M-80 version and copied to an MS-DOS formatted disk (using the format translating program Uniform) can be loaded and run in the

MS-DOS version of MultiPlan. When I tried to go the other way, from MS-DOS to CP/M 80, it did not work. I got the message that the file is "not in spreadsheet format."

As mentioned above I have copied the MS-DOS file COMMAND.COM to my MultiPlan "program" disk so that I can exit the spreadsheet and return to MS-DOS with no disk swapping.

Running the RAM Disk

I have done little more than copy the distribution disks for MS-DOS and CP/M-86 and, of course, run the RAM Disk. This is very simple. Both the MS-DOS "Load" disk and the CP/M-86 software distributed by SWP contain the RAM Disk software.

The RAM Disk is activated by typing `RAMDISK` ↵ and answering questions about what to "name" it, whether or not to "clear" its directory (you normally type `Y`, for yes), and a couple of questions about the choice of its location in memory. If you choose the default value with a Return, it loads just below the CCP (Console Command Processor portion of CP/M).

You can configure the RAM Disk to any drive letter (from A to P) you want. It provides 252K of formatted space. If you choose to name it Drive A, then the two existing floppy disk drives become B and C and the system tracks are copied to the RAM Disk. Any subsequent "warm boot" (Control-C) reads the system from the RAM Disk.

If you name it "Drive B" then, existing drive A remains A, and former drive B becomes drive C.

If you name the RAM Disk Drive A, the system tracks are automatically copied to the RAM Disk so that "warm boots" can be performed directly when you type Control-C with no mechanical running of either of the floppy drives.

We ran WordStar® from the RAM Disk with good results. With WordStar and the text file both on the RAM Disk, execution speed was somewhat improved. For example, scrolling from the beginning to the end of a 65K text file was reduced from 45 seconds on a floppy disk drive to 10 seconds on the RAM Disk. The real payoff in time saving should come, however, in data transfers where there is a lot of input/output and disk read/write operations.

The RAM Disk is useful for running any CP/M-80 application program. dBASE II™ operations, for example, would be improved. It also makes it much easier to copy a disk with a two-drive Osborne, using the Ram Disk as drive A.

All things considered, the CO-POWER-88 board is worth the price. It does turn your Osborne 1 into an MS-DOS or CP/M-86 computer, while keeping its CP/M-80 personality, making your computer suitable for running both 8-bit and 16-bit programs. It may be just what you need to keep your feet in both the 8-bit and the 16-bit worlds.

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